This section describes the *design* of the system and its subsystems, as opposed to the *requirements* for the same. (Requirements define what a system, subsystem, or component shall do. Design documents describe how the designer chooses to implement the requirements, and why the implementation was chosen.)

4.1 Design overview

This section gives an overview of the system design: the functionality, hardware, and software used to implement the system.

4.1.1 Functional description

The system provides four major functions, as outlined in section 2.2: user interface, image generation, assist with maintenance, and external interfaces.

4.1.1.1 The user interface

The user interface consists of a workstation, with a wide range of system functions, and a set of hardware controls, located to ensure convenient workflow.

The workstation is used for the majority of UIF tasks: scheduling, information entry, protocol management, scan management, and image management. The design intent is to provide a single workstation, with flexible and familiar graphical interface, that implements all the necessary tasks. The range, complexity, and flexibility needed by these functions requires an interface with similar complexity.

There is also a limited set of hardware based user controls, located so that the operator can reach them when they are needed. These functions include patient handling controls (landmark, table motion, etc), scan controls (start, stop, pause), patient intercom controls (volume), and emergency system controls (emergency stop and emergency off). There is a grouping of controls accessible in the scan room, and another grouping on the operator workstation. Emergency off and stop controls are distributed throughout the site to provide ready access.

4.1.1.2 Setup and generate images

The image setup and generation function is a highly automated function. It begins when the operator completes the scan prescription, and initiates the scan. The design intent is for the system to automatically self-calibrate to the patient, execute the scanning process, process the data from the scan, and place images or other data into the system's database. The operator is intended to have minimal involvement with this function, freeing them for other tasks.

4.1.1.3 Assist with maintenance

System maintenance is a complex, expensive process. The service personnel require a high level of training, and experienced field engineers (FEs) are a valuable resource. System downtime, for either regularly

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scheduled maintenance or unscheduled repairs, has adverse effects for both system owners, operators, and patients.

Therefore, the system is designed to assist in the maintenance process and guide FEs in diagnosis and post-repair testing. Reference [4], the *MR Serviceability SDD*, gives more detail on the system's features that support system maintenance and service.

4.1.1.4 Provide external interfaces

The system provides several external interfaces. Most of these interfaces provide access to fundamental signals so they can be viewed by design engineers. These include access to the InView signal (when the receiver is digitizing data), the 10MHz clock, exciter unblank, combined unblank and MNS clock. The system also provides the ability for a pulse sequence to output a signal at a specific time so that external devices can be in sync with the scanner. These connections are labeled DABOUT.

DOC2061516, the *Non-Standard System Interface Design Document*, has a wealth of information on these interfaces.

4.1.2 Block diagram

The block diagram below is an overview of the system implementation. It shows the main subsystems, and key physical and functional relationships.

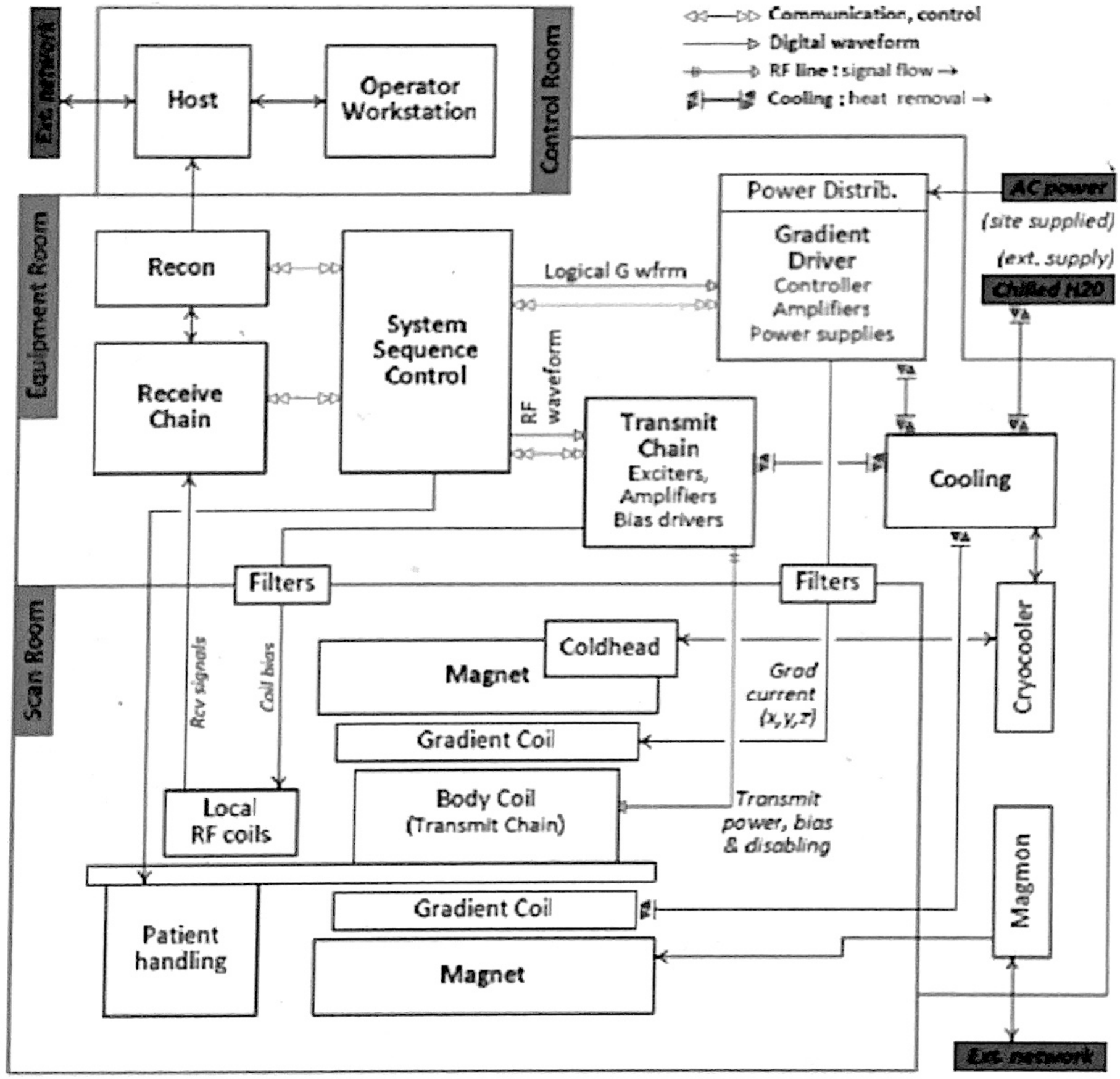

Fig. 4–1: System Block Diagram

4.1.3 System configurations

This section outlines GEHC Premium systems – MR450, MR450w, MR750, MR750w, and Rio.

System	MR450	MR750	MR450w	MR750w	"Rio"
B0 field strength, T	1.5	3.0	1.5	3.0	3.0
Patient bore size, cm	60	60	70	70	70
FOV (R x Z), cm	48 x 48	48 x 48	50 x 45	50 x 45	?
Magnet type (series)	R (LCC)	W (G3)	HM (DVw)	UM (3TLC)	UM or W
Shim	Supercon	Supercon	Passive	Passive	Passive

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System	MR450	MR750	MR450w	MR750w	"Rio"
Gradients	--	--	--	--	--
Coil	XRMB	XRMB	XRMw	XRMw	HRMw2
Driver	XGD/XG2	XGD/XG2	XGD/XG2	XGD/XG2	RGD
SR (clinical), T/m/s	200	200	120	120	150
Gmax (clinical), mT/m	50	50	33	33	70
RF subsystem	--	--	--	--	
RF amplifier [1]	8103	8135	8103	8137	8137
Rated Power	16KW	35KW	16KW	15KW x 2	15KW x 2
Body coil	16 run	16 run BP	16 run	16 run HP	16 run HP

Notes:

(1) "81" series amplifiers are from Analogic.

(2) BP = bandpass, HP = highpass.

4.1.4 External interfaces

4.1.4.1 Isocenter location

System isocenter is defined as the center of the imaging volume. In measurable terms, this is where the three gradient "iso-planes" meet at a single point, which in turn determines the center of the imaging space.

The system isocenter is located roughly 107cm above the finished floor.

(Note that there are additional specifications detailing the spatial relationships between the magnet and environment, but these are outside the scope of this document.)

4.1.4.2 DICOM

DICOM (Digital Imaging and Communications in Medicine) is a standard for handling, storing, printing, and transmitting information in medical imaging. The system can read and write images that conform to the DICOM standards.

In general, the system is intended to conform the applicable DICOM standard.

The system's DICOM implementation creates DICOM objects for Secondary Capture Image (screen save) in both Gray Scale and Color; for Gray Scale Softcopy Presentation State (GSPS); and for Structured Report (basic, enhanced, and comprehensive).

The DICOM format allow for additional special purpose entries. The system's DICOM implementation accepts DICOM objects for CT images, PET images, RT structures, GEMS PET Raw information.

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4.1.4.3 Filming

4.1.4.4 PACS

4.1.4.5 External triggering

4.1.4.6 Local coil interfaces

The system provides two types of user accessible local coil ports (user accessible connectors).

4.1.4.6.1 Type "A" connectors

"A" connectors are multifunction ports, including 1) transmit power, 2) 8-channel receive, 3) DC power, and 4) multichannel bias.

The "A" port includes monitoring pins, used for detecting connection or disconnection of local coils.

4.1.4.6.2 Type "P" connectors

"P" connectors are multifunction ports, including 1) multi-channel receive, 2) DC power, and 3) multichannel bias.

The "P" port includes monitoring pins, used for detecting connection or disconnection of local coils.

The "P" ports include a mechanical device to prevent local coils requiring transmit power from being connected to them.

4.1.4.7 Insite / OLC

4.1.4.8 Accessory qualifications

4.1.4.9 Applications

4.1.4.9.1 FUS

4.1.4.9.2 Brainwave

4.1.4.9.3 CADStream

4.1.5 Subsystem overviews

This document's coverage of subsystems is organized into several sections. The organization of the content -- what information goes where -- is guided by several considerations. In priority order, functions and components are grouped into subsystems by 1) functional association, 2) physical association, and 3) other considerations.

As an example, all gradient related components default to the gradient subsystem – a functional association. However, some components are related to several groups; the RF shield serves to isolate the gradient and RF fields. Here the second consideration places it in the gradient subsystem, as it is usually built into the gradient coil.

1. Magnet subsystem

The magnet subsystem creates the *B0* static magnetic field, and contains the magnet and related components. This includes 1) the magnet, 2) the coldhead and cyrocooler, 3) the magnet monitor, 4) any

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shimming components, passive or active, 5) the Magnet Rundown Unit and 6) mounting components, including vibration and seismic mounts.

2. Gradient subsystem

The gradient subsystem creates the G_x, G_y, and G_z gradient fields, and contains the related components. This includes 1) the gradient coil, which includes the RF shield, 2) the gradient amplifiers and power supply, 3) the gradient controller, 4) the gradient filters, 5) the gradient cables, busbar, and coil interface, and 6) hoses and fittings for cooling flow. Its main interfaces are 1) command and waveform information from the SSC, 2) AC power from power distribution, 3) cooling from the cooling subsystem, and 4) mechanical interfaces to the magnet and TXC's body coil.

3. Transmit chain (TXC)

The transmit chain creates the *B1* RF transmit field, and contains the related components. This includes 1) the exciter, 2) the RF amplifier, 3) transmit RF power monitoring, 4) transmit cables, 5) RF switchgear, 6) the RF whole body coil, and 7) the transmit and coil bias drivers.

4. Receive chain (RXC)

The receive chain processes the received MR signals, and contains the related components. This includes 1) preamplifiers, 2) demodulation and digitization signal paths, and 3) time-based filtering.

5. RF coils

RF coils include all receive-only RF coils, and all T/R coils, except the RF whole body coil. This includes the antenna structure, biasing and blocking networks, on-board preamplifiers, housings, cabling and connectors.

6. Patient Handling

The patient handling subsystem includes the scan room components accessible to the patient or operator, with the exception of RF coils. This includes 1) the operator user-interface, 2) physiological data acquisition, 3) patient tables and associated controls, 4) in-room displays, and 5) the patient comfort system (intercom, bore lights, and bore ventilation).

7. Recon

The reconstruction subsystem includes the hardware and software components used for converting MR data into reconstructed output, such as images, spectroscopy signals, etc.

Other areas that are not yet included in this document are:

Scan & Equipment Room, Op Workspace, Thermal management, Power Distribution, Cooling , System Sequence Control, System Tools, Pulse Sequence, Host Software, Coils, Diagnostics, DICOM and Annotation, Host Tools, Visualization, and Error Messages, Documentation, and Labeling

4.2 System level design considerations

4.2.1 The Four "C"s

The system design includes a wide range of tools used to support the "Cs": corrections, calibrations, characterizations, and compensations.

- **Corrections** address imperfections in a system or component design. A correction is applied the same way to all systems of the same design.
- **Calibrations** (or cals) assist in setting up or adjusting the system, either to improve performance or reduce variability between systems. *A calibration affects a system's operation or performance by changing some aspect of system performance, based on measurement of that specific system.*
- **Characterizations** measure system performance. These tools generally aid in troubleshooting problems or understanding differences between systems.
- **Compensations** are real-time adjustments performed to address manufacturing variations. A compensation is based on the characteristics of a specific system.

Most tools are integrated into the system software, with a degree of automation for efficient use and consistent performance.

Item	*Correction*	*Calibration*	*Compensation*	*Characterization*
DAQ acq & RF delays	*			
gradwarp	*			
Passive Shim		*		
TRDD		*		
RF Amp cal		*		
UPM cal		*		
EPIWP				*
Grafidy3 ECC			*	*
Quad DD		*		
LVShim		*		*
HOEC			*	*
Isocenter (DQA)		*		
Gradcal (DQA)		*		
System Gain		*		
B0 Drift			*	*
SPT (SNR,Stab,CNT,ECchk)				*
EPT		*		*
Probe Tune/SNR (P,S)		*		*
RFA				*
HSS				*

4.2.1.1 Corrections

Corrections address imperfections in a system or component design. Unlike calibrations, which will vary between systems of the same design, *a correction is applied the same way to all systems of the same design.* For example, the *gradwarp* correction compensates for gradient coil non-linearity; the same *gradwarp*

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corrections are used in all systems using "XRMw" gradient coils. However, these corrections would be different than those used on "XRMB" based systems.

4.2.1.1.1 DAQ and RF delays

The time alignment between RF, gradient, and data acquisition (DAQ) waveforms is critical. The pulse sequences assume that the waveforms are synchronized. However, the time delay thru the RF, gradient and DAQ subsystems is generally not the same, resulting in timing errors between the pulses from the different subsystems. A timing correction, using two adjustable parameters is used to align the three subsystems' outputs.

The start of sequence marker is used as a reference. The gradient waveform timing is not adjustable. The `psdGradWait` parameter is used to adjust the DAQ timing, relative to the sequence marker and gradients. The `psdRfWait` parameter adjusts the RF pulse timing. These parameters are kept in `/w/config/GradientConfig.cfg`.

The config values, in µs, are added to the normally calculated time positions for the RF and DAQ waveforms. Only positive values are allowed for `psdRfWait` and `psdGradWait`, so the waveforms can only be delayed, not "advanced".

These config values are determined at system design time by testing. They vary by system configuration, but are not set on a system-by-system basis.

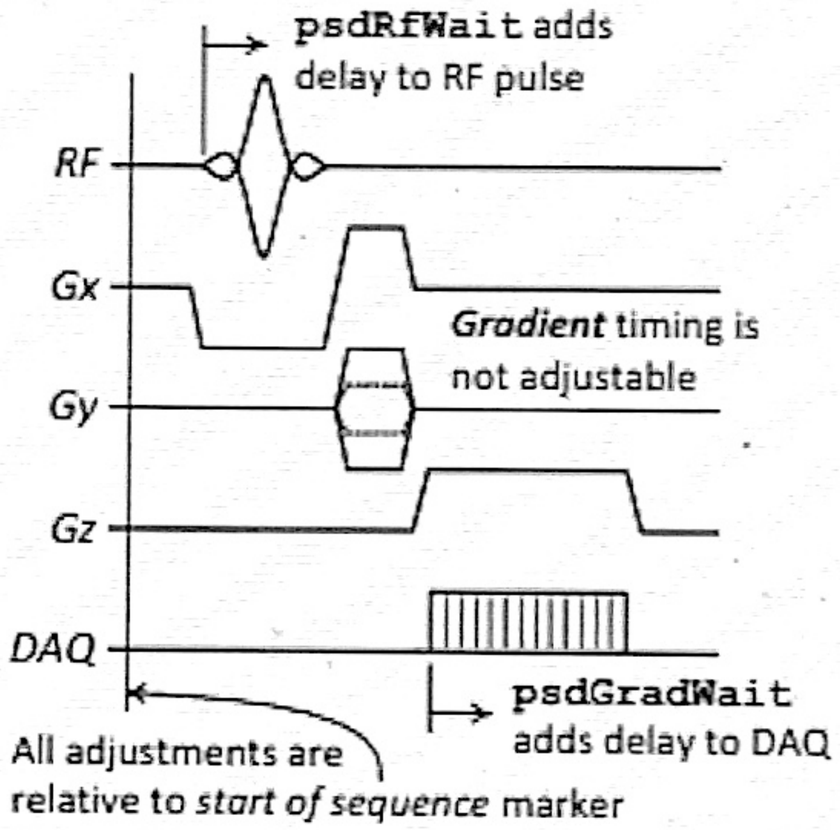

Fig. 4–2: Waveform timing adjustments

Historically, the "two delays" approach, with only positive values, made sense since the gradients usually had the longest inherent delays (the first systems had gradient bandwidths of <2KHz.) Therefore, delay was added to the other waveforms to realign them with the slower gradient waveforms.

Note that using only two parameters in the design implies that the timing of the X, Y and Z gradients are the same. This is not necessarily true but in practice, the timings are within 2usec.

4.2.1.1.2 Gradwarp

The *gradwarp* correction reduces image distortion by applying geometric corrections to images, based on design time characterization of gradient spatial non-linearity. (Refer to §3.7.1.4 *Gradient FOV* for a discussion of gradient non-linearity).

Fig. 4–3 shows a FFD (full field distortion) phantom, in the *xz* plane. Each square is 5cm, so this is a 30x40cm grid.

In this example, *gradwarp* uses two-dimensional stretching, or remapping. The left image has *gradwarp* turned off; the right image has it applied.

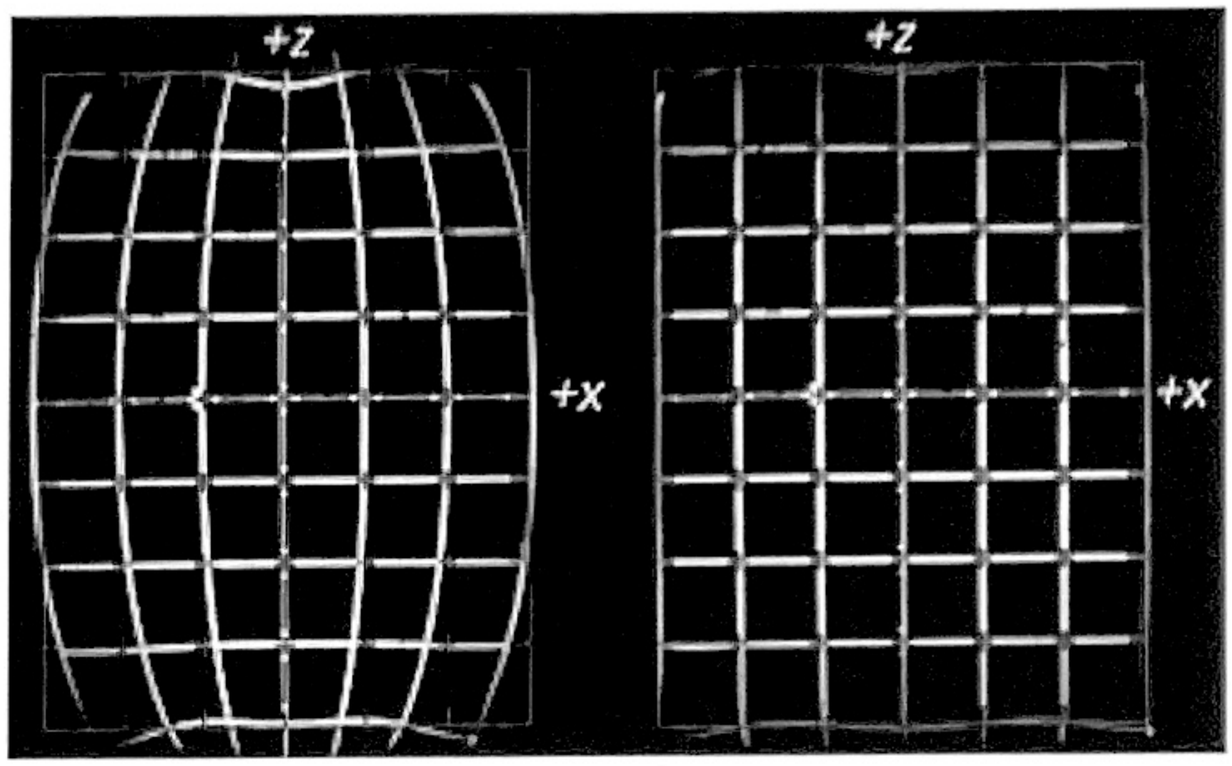

Fig. 4–3: Effects of *gradwarp*

The remapping functions (or coefficients) are unique to a gradient coil design, but apply across all systems using that design. All systems with XRMB coils use the same coefficients, which are different from those used in all XRMw based systems.

(The images shown are from a 2002 design, the TRM whole body coil.)

Gradwarp works well for correcting geometric distortion. However, there are limitations. For example, the outside corners of the phantom are missing in the no-*gradwarp* image; this is because the actual slice is not a flat plane, but curls up near the corners, missing the phantom, which *is* flat. 2D image remapping cannot restore the missing corners.

There are two types of *gradwarp* corrections: 2D and 3D. The 2D method applies the correction for each slice in one spatial direction before applying the correction in the second direction. The 3D method applies the correction in all directions simultaneously. For further information, refer to the SWDD for 3D gradwarp (DOC0681574).

Three dimensional (3D) *gradwarp* requires true 3D image datasets. In addition to the 2D distortion shown above, the image "plane" is generally not flat, especially for slices that are not near isocenter. Therefore, the remapping must use data from a slab surrounding the image requiring 3D correction; the original image won't have all the information needed for 3D *gradwarp*.

4.2.1.1.2.1 Coil characterization and coefficients

During a gradient coil's design cycle, the gradient fields are calculated using Biot-Savart methods from the gradient coil winding patterns. The calculated fields are decomposed into spherical harmonics (see Appendix B3, *Spherical Harmonics*). Ideally, gradient fields have odd on-axis symmetry, and even off-axis symmetry; therefore, the gradient fields are primarily composed of only a few terms.

Specifically, the dominant terms for each gradient axis are:

x-axis: $l = 1,3,5,7, \ldots$ with $m = 1$
y-axis: $l = 1,3,5,7, \ldots$ with $m = -1$
z-axis: $l = 1,3,5,7, \ldots$ with $m = 0$

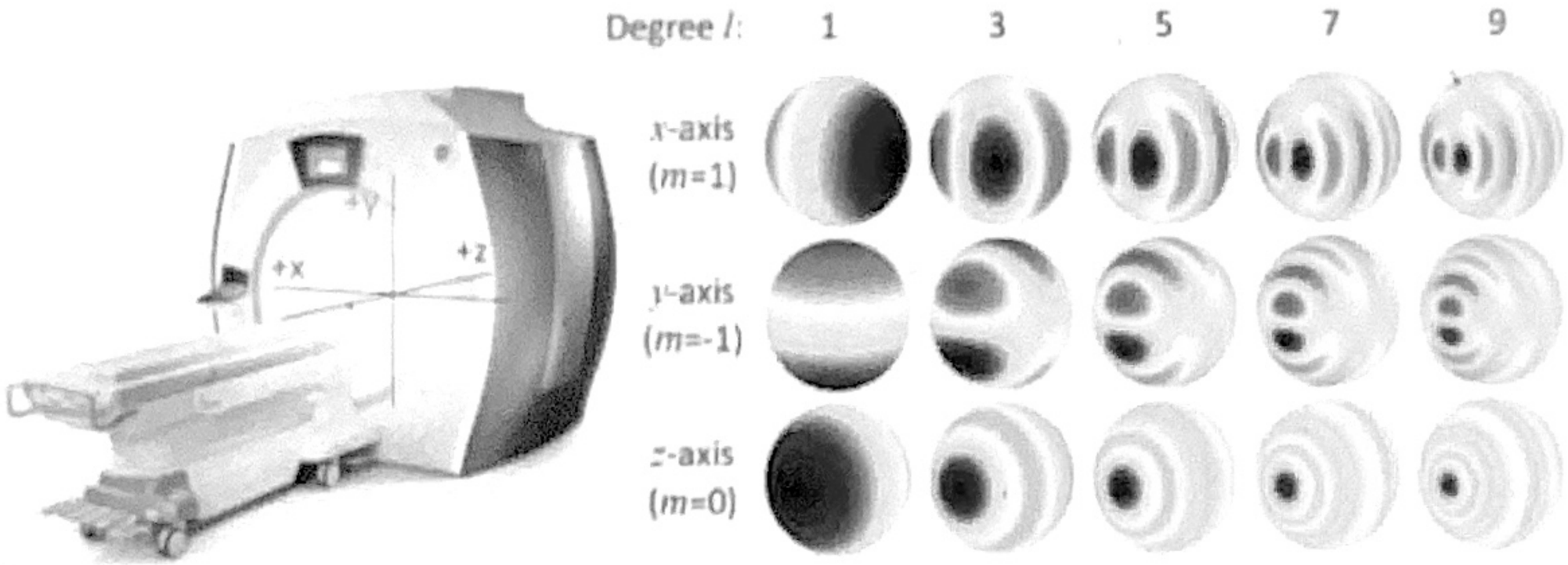

Fig. 4–4: Spherical harmonics used in *gradwarp*

Note that Fig. 4–4 has the spherical harmonic's xyz axes rotated to match the scanner's axes. (Appendix B3, *Spherical Harmonics*, has a different orientation, with the z-axis pointing up).

Inspecting the figure shows that the spherical harmonics used in gradient field correction will always be zero on the iso-plane normal to an axis. For example, all the harmonics used in the y-axis characterization are zero on the xz plane.

For pre-HRMw gradient coils, the coil's nonlinearity is characterized using the on-axis third, and fifth degree harmonics for each axis. The initial decompositions were used to remap (apply geometric correction to) images of a large gridded phantom (the FFD phantom). These images were used to refine the decompositions; typically, the 5th order Z term was adjusted to compensate for a significant 7th order term (which was not supported in the *gradwarp* algorithm), thus improving the overall correction.

The HRMw gradient coil's *gradwarp* uses the 3rd, 5th, 7th, and 9th order terms to improve the spatial correction accuracy. Even with the improved correction capability compared to DV systems, it was shown that there are additional small but non-negligible terms (refer to DOC1912383). The ability to correct for these terms has not been implemented.

The correction coefficients are stored in `/w/config/gw_coils.dat` and read by the reconstruction subsystem for *gradwarp* as well as the host subsystem to perform "reverse gradwarp" for 2D acquisition slice positioning.

4.2.1.2 Calibrations

Calibrations (or cals) assist in setting up or adjusting the system, either to improve performance or reduce variability between systems. *A calibration affects a system's operation or performance by changing some aspect of system performance, based on measurement of that specific system.*

Most cals require use of some system performance measurement to guide adjustment. For example, a measurement of eddy currents is required to complete eddy current calibration.

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The *overall calibration strategy* is to reduce the observable performance difference between systems to below the level of customer concern. From the customer perspective, all systems of a particular design should give equal performance; any MR750w system should perform as well as any other.

Order of calibrations

Most of the calibration procedures are image-based; system imperfections and "not-yet-calibrated" subsystems may impact the results. Therefore, the order in which cals are done affects the quality of the results and the efficiency of the process. This section covers the cals in the order that they are usually completed.

4.2.1.2.1 Passive shimming

Passive shimming consists of placing small ferromagnetic pieces (patches or pellets) inside the magnet to improve the B0 homogeneity. It is an interactive process: measure, add or remove material, and repeat. The process completes when the overall homogeneity is within acceptable limits.

The tool guiding the p-shim process is a PC-based software package. Created and owned by the Florence Engineering team, it runs on the Service Engineer's laptop. (As such, it is not part of the software release.) In general, the magnetic field data that is input to this tool is generated by a field camera. An array of magnetometers is placed in the magnet bore, measuring the field strength at an array of specific points in the bore. The patient cradle must be out of the bore to collect the full array of field points.

There are two major advantages of this tool. First, the tool does not require a functional MR scanner. Since the MR scanner is not used, the only requirement is a ramped and stable main magnetic field. Second, the tool output is not susceptible to calibration status or MR artifacts. The major disadvantage of this tool is that it requires an empty bore and a dedicated shim camera and in the case of MR750w and Rio, a dedicated shim tray extraction tool.

4.2.1.2.2 TRDD calibrations

The TRDD bias calibrations measure and set up limits on the bias driver circuits (located in the RF driver module) that are used to drive the dynamic disabling (DD) networks. The calibration increases the chances of detecting a hardware failure by removing the variation in dynamic disable circuit loads between systems.

To calibrate, the driver circuit is put into current sink mode and the voltage of the load is measured. A limit is set above this measured value such that open circuits (in the dynamic disabling networks of either the hybrid or RF body coil) can be detected.

4.2.1.2.3 RF amplifier calibration

RF amplifier calibrations adjust the RF amplifier gains so that the specified full RF power is produced at maximum system transmit gain.

The *RF Amp Cal* tool measures the output power of the RF amplifier, which guides the FE to adjust the RF amplifier gain to set full scale power. The Amp Cal tool consists of a pulse sequence (*ampcal*) that provides trapezoidal pulse waveforms, and a Power Measurement kit that includes a RF wattmeter, a dummy load, and appropriate cabling. The *ampcal* sequence ensures that the exciters are set to produce full scale signal

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when TG=200; the FE uses a manual RF amp gain adjustment the RF amp's output power to full scale (which varies by platform.)

The RF amplifiers have a body mode (high power) and a head mode (low power). These modes require separate calibrations. In the current designs, there is some interaction between the two modes. Due to this interaction, the body mode is calibrated before the head mode (implying if the body mode calibration is changed, the head mode needs to be checked).

4.2.1.2.4 UPM calibrations

The UPM (Universal Power Monitor) calibrations set the gain for each active UPM channel, based on a full- scale pulse from the RF amplifier.

A square-wave RF pulse is applied and the UPM's gain adjusted until the channel under test records the correct amplitude.

4.2.1.2.5 Quadrature calibrations (Quad Cal)

This calibration applies to systems with dual drive transmit capability, and adjusts the amplitude and phase balance (ΔA and ΔP) between body coil CH1 and CH2 to optimize the B1 field quadrature.

The quadrature calibration design applies the same RF waveform to both channels, while allowing a limited range of phase difference and amplitude adjustments between channels. If a ΔA or ΔP setting outside this limited range is needed, the calibration fails.

Typically, ΔA is described in "counts", with 1 count = 0.1dB. ΔP is described in degrees.

4.2.1.2.5.1 DDQuad (Gen 1 Quadrature Calibration)

The **Dual Drive Quadrature Tool** implements the first generation of quadrature calibration. It is variously called "DDQuad", "Dual Drive Quadrature", "Legacy Quad Cal", or "QC-I".

Three images are obtained: the first using both transmit channels, the second and third using only CH1 and CH2, respectively. The first image (using both channels) checks for adequate image SNR and uniformity, and provides an appropriate TG setting. The differences in the second and third images are used to determine amplitude and phase settings needed to make the two images identical. Algorithm details can be found in DOC1519360, *DV25 DD Quadrature Tool SWDD*

DDQuad only calibrates transmit quadrature. DDQuad is being phased out, and replaced by QuadCal (below), starting with the Rio program.

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4.2.1.2.5.2 QuadCal (Gen 2 Quadrature Calibration)

The **Quadrature Calibration Tool** is the second-generation quadrature calibration. It is generally called "QuadCal" or "QC-II". In addition to new transmit quadrature calibrations, it has the first implementation of receivè quadrature adjustments.

The transmit side uses the interaction between the B1+ field and a large, highly dielectric, spherical phantom to adjust transmit quadrature. Since the phantom is spherical, a quadrature B1+ field will produce round B1 shading patterns in the image; the ΔA and ΔP adjustments are used to adjust the roundness of the shading patterns.

The resulting characterization is used to predict ΔA and ΔP for an ideal round shading pattern.

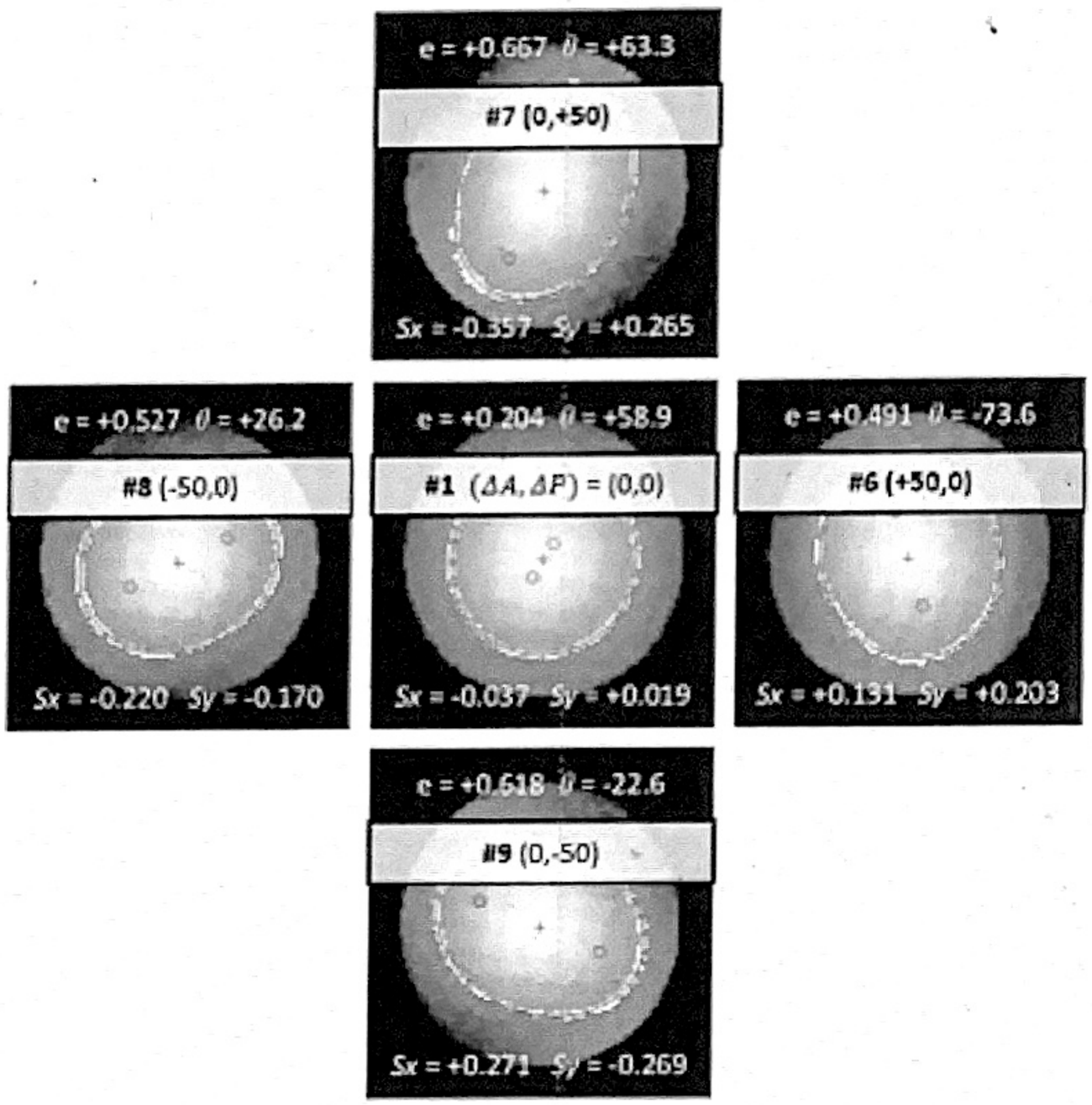

Fig. 4–5: Transmit side QC-II

The receive calibration uses the complex quotient of receive Ch1 and Ch2 SpinEcho images to quantify the residual magnitude and phase corrections needed. This method assumes that the Tx calibration is complete, and that all remaining eccentricity is due to a non-quadrature receive state.

The QC-II transmit side design is described in DOC1803495, *Elliptical Quadrature Calibration*. This document provides an overview of the rationale, method, and algorithm used for transmit side QC-II.

Details on the QuadCal Tool rationale, methods, and algorithms can be found in DOC2055366, Quadrature Calibration Tool SWDD.

4.2.1.2.6 LVShim

LVShim (large-volume shim) is an image-based tool used to adjust current through dedicated coils to increase the static magnetic field homogeneity. The tool uses a large phantom (split into three pieces to control the maximum weight of each piece) to cover a large area in the bore. In order to provide this function, the LVShim tool also provides the ability to characterize the homogeneity of the static magnetic field.

The tool operates slightly differently depending on the magnet type, passive (wide bore) or superconductive shims (G3, LCC). For superconductive shim magnets, this tool provides shim capability. It controls image

acquisition, image processing and then indicates updates to the currents in the superconductive shim windings currents to alter the shim. The field is decomposed into spherical harmonic components and the specifications for the field are in terms of these components. A separate mode of the tool sets the initial values of current through the gradient coil windings for the first order components.

For passively shimmed magnets, the LVShim tool is only used to characterize the static magnetic field as well as setting the initial values of current through the gradient coil windings for the first order components. The Passive Shim tool is used for the main shimming process. The tool defines the acceptance criteria as peak field deviation over several volumes.

LVshim requires no additional hardware except the phantom when measuring the field or setting the default current through the gradient coil windings. When used for shimming superconductive shim magnets, a power supply and shim lead is required to apply currents.

4.2.1.2.7 Isocenter

The *isocenter* calibration adjusts the distance the table moves when travelling from the landmark position to the scan plane. This calibration changes `Mrconfig.cfg` parameter `isoVectorZ`, which is the table's landmark-to-isocenter travel distance, in 0.1 mm units.

This process only calibrates the z-axis isocenter. There is no cal for x- or y-axis isocenter.

The isocenter calibration is an image-based process. It is part of the DQA tool, and uses the DQA-III phantom. The DQA-III phantom includes a block with an angled cut thru it, aligned with an external mark on the phantom. After landmarking on the phantom's mark and advancing to the scan plane, an axial image is obtained. By checking where the block's angled cut intersects the image plane, the location of the phantom's landmark relative to imaging isocenter can be determined. Any distance reflects an error between the actual table motion and the actual distance, so it is added to the current `isoVectorZ` value and the calibration repeated.

Note that for this calibration to be independent of relative height, the alignment light must be properly adjusted such that it is vertical (perpendicular to the top of the patient handling system).

4.2.1.2.8 Gradcal

The *gradcal* calibration sets the overall gain of the gradient subsystem, from digital command to gradient field amplitudes G_x, G_y, and G_z. This calibration changes `GradientConfig.cfg` parameters `xfull`, `yfull`, and `zfull`, which sets the digital scaling of the full-scale gradient field. If `xfull` is set to 30733, the waveform generation system will output 30733 to obtain a full scale (as defined in the GradientConfig.cfg file) waveform on the *x*-gradient.

The calibration is part of the DQA tool. It uses the DQA-III phantom, which has internal slots at 10.00 cm (far edge to far edge distance). The distance between the slots is measured on an image, and the `xfull`, `yfull`, and `zfull` values adjusted accordingly.

Note that three new parameters were added to the GradientConfig.cfg file for the Rio product. These three parameters, `xfull_default`, `yfull_default` and `zfull_default`, were added to support the two system configurations, Premier and Premier XT, that differ in provided gradient capability but not hardware (hardware components are the same). These parameters represent the nominal gradient calibration values

for the specific configuration. These values are used to ensure appropriate starting values are used for the calibrated gradient field gain when converting between configurations.

4.2.1.2.9 System gain

The *system gain* calibration sets an image gain factor used in the image reconstruction process. The purpose of this calibration is to adjust the default image intensity such that they are similar comparing images from the same type of scanner.

There are two gain factors: head and body. The tool operates by running a specific protocol using either the quad head coil or body coil with the TLT phantom and adjusts the gain factor such that the average phantom image intensity is within a specified range. Note that since this factor is used to adjust all of the pixel values, there is no impact to the resulting SNR value obtained from the image.

For simplification, Rio consolidated System Gain calibration into the SPT SNR tool. The *"SNR and System Gain calibration tool"* uses the legacy SNR protocol to do both the SNR analysis and system gain calibration. As a result, system gain target intensities were redefined. It has been confirmed that legacy and new system gain calibration will lead to same calibrated system gain (recon scale factor) values.

For more details refer to DOC2002976, the *SNR and System Gain SWDD*.

4.2.1.2.10 EPT

EPT (Echo Planar Test) is a collection of tools to calibrate and characterize the system for echo planar imaging. Due to the data acquisition order, EPI is more sensitive than most sequences to small eddy currents. Note that since EPI's inception, improvements to the system phase correction algorithms have made the calibrations less critical.

a) **White Pixel**: Incorporation of the EPIWP tool (see section 4.2.1.3.1) for convenience. EPT offers the tool for both head and body modes.

b) **B-zero Dither**: A calibration to account for non-zero phase offsets as a function of echo spacing.

c) **Group Delay**: A calibration to align the center of the signal echo with the center of the data acquisition window as a function of gradient axis.

d) **B-zero Image**: Characterization of the image ghosting running EPI sequences varying the imaging plane and echo spacing.

4.2.1.2.11 Probe Tune

Probe Tune is a calibration to align the center of the signal echo with the center of the data acquisition window as a function of gradient axis. There are two calibration tools which apply to the two types of probe sequences: Probe_S and Probe_P. Probe_S uses a STEAM pulse sequence, collecting a stimulated echo. Probe_P uses a PRESS sequence, collecting a spin echo. The calibration values are stored in `/usr/g/caldir/probes_cal.log` and `/usr/g/caldir/probep_cal.log`, respectively.

4.2.1.3 Characterizations

Characterizations measure an aspect of system behavior, to help understand a problem or compare one system to another.

4.2.1.3.1 EPIWP

The EPIWP (Echo Planar Imaging White Pixel) tool measures if the system produces high-intensity signals in k-space not related to the RF signal. The Echo Planar Imaging protocol is sensitive to this artifact and is able to create the artifact effectively when the proper conditions are present. The tool operates with no RF.(no phantom required), collecting and displaying k-space data of EPI sequences that vary in echo spacing and axis orientation.

The usual conditions that cause white pixels are 1) metal-metal contacts that are in the gradient field and not secure or 2) a failed gradient coil.

In imaging space, how 'white pixels' show up in images depends greatly on the type of protocol and when the high-intensity data was collected related to the center of k-space, but often shows up as the obviously named "corduroy" artifact similar to Fig. 4–6.

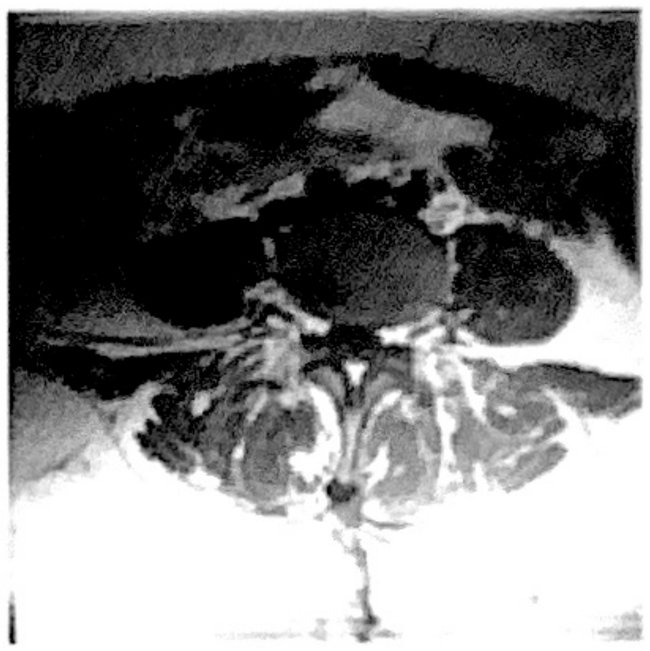

Fig. 4–6: A corduroy artifact

4.2.1.3.2 SPT

SPT (System Performance Test) is a collection of tools for characterizing a range of system performance parameters. These parameters include SNR (signal to noise ratio), coherent noise, eddy currents and signal stability. SNR and stability measurements can be collected with either the quad head RF coil or the RF body coil.

- **SNR**: The SNR measurement uses an FSE protocol to characterize the signal to noise of the system. Details can be found in DOC2002976. Note that as mention in section 4.2.1.2.9, when used during initial system calibration, the system gain calibration is included on Rio systems.
- **Stability**: Stability measurements are separated into orientation (axial, sagittal and coronal) as well as pulse sequence. The FSE or GRE pulse sequence may be used. The repeatability of the signal from multiple slices is characterized. In general, the FSE sequence indicates the performance of the RF chain while the GRE sequence indicates the performance of the gradient chain. Details can be found in DOC2056787.
- **Coherent Noise**: The CNT digitizes signal without using RF energy. This tool is designed to characterize extraneous energy being picked up by the quadrature head RF coil in the frequency range that you are receiving. Details can be found in DOC2058149.
- **Eddy Current Checks**: A tool similar to the HOEC tool is provided to characterize B0 and long eddy currents. Like HOEC, it is an image-based tool. This tool was written to provide the capability to perform a rough check of eddy current compensation during Preventative Maintenance without the need for the grafidy kit. Details can be found in DOC2057529.

Starting with the Rio program, SPT was split into standalone Matlab tools controlled by the System Tools Controller (STC, detailed in DOC2002974).

4.2.1.3.3 RFA

RFA, for RF Analysis, characterizes the RF transmit chain's performance. The system's receive capabilities are used to measure the output of the transmit chain. (This requires a special purpose pulse sequence to allow simultaneous transmit and receive.) RFA can characterize the TXC for

1) amplitude and phase transfer function,
2) the amplitude and phase linearity,
3) interpulse stability,
4) transient behavior.

The tool has the capability to run in different modes to be able to test the output of the exciter, RF amplifier and RF coil. The RF coil mode has not been usable since the receivers were placed on the side of the magnet.

4.2.1.3.4 HSS

HSS, for High Speed Stability, characterizes B0 field instabilities. A small MR sample, with matching RF coil, is used to produce a series of short FIDs, one every 2.5ms. The frequency as a function of time is determined based on the phase slope of each FID. These measurements are analyzed to characterize the B0 field stability, in both the time and frequency domains. The sample can be moved to different bore locations, or multiple samples used at different locations, to determine the spatial characteristics of the B0 field instability.

4.2.1.4 Compensations

4.2.1.4.1 Eddy currents

Pulsing the gradient fields induces currents in conductors near the gradient coil. These *eddy currents* (or ECs) create an additional magnetic field, which superimposes onto the system's gradient and B0 fields. Similarly, pulsing the gradient field will induce mechanical motion in nearby conductors; if the conductors are carrying current, the motion changes the magnetic field created by the current, producing eddy current like effects.

Eddy current's magnetic fields vary in both time and space. Over time, eddy currents decay exponentially, sometimes with oscillation, with decay time constants between several microseconds and several minutes. Spatially; eddy current fields have significant zero and first order components; the zero order components appear as a change in the B0 field, while first order components look like a change in the gradient fields. Higher order eddy current components have been measured, but have not been found to be as significant as the zero and first orders.

A highly integrated service tool to measure eddy currents and set up EC compensation, *grafidy*, is provided with the system. The *grafidy* tool consists of a measurement kit (including multiple transmit/receive MR probes and a fixture to hold them), and pulse sequence (which is also called *grafidy*) to drive the gradients and acquire measured data, and an analysis program (to process the acquired data and update system eddy current compensation calibration files.) This tool compensates for zero and first-order eddy currents with a wide range of time constants.

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Performance limits are placed on the uncompensated eddy current field, and the residual EC fields remaining after compensation. These limits apply to zero and first order EC fields. For DV systems, there are no limits in place on higher order EC fields. With the improvement in the gradient coil design tools, the Rio system (HRMw) included a post-compensation requirement for a given magnet.

Eddy currents are primarily an interaction between the gradient coil's residual field (what extends beyond the outer diameter of the GC) and the conductive elements in the magnet. A GC outside a magnet does not accurately reflect the ECs seen with the same GC inside a magnet. Therefore, the environment surrounding the GC must be specified when specifying uncompensated ECs. Uncompensated ECs are reduced by GC designs that lower leakage fields, or magnet designs that use less conductive materials near the GC.

4.2.1.4.1.1 Eddy current terms

The overall eddy current behavior of the system is characterized as a sum of many terms. Each term has

- a stimulus gradient (one of the x-, y-, or z-gradient fields),
- a response component (either the B0 field, x-axis gradient, y-axis gradient, or z-axis gradient),
- a time constant (a real number for exponential ECs, or a complex value for oscillatory ECs),
- an amplitude (usually expressed as a percentage of the stimulus field), and
- a frequency if the response is oscillatory in nature.

A shorthand notation shows the stimulus and response for an EC term. An x→y eddy current has an x-gradient stimulus, and a response affecting the y-gradient field; this is referred to as "x to y". A z→B0 eddy current has a z-gradient stimulus, and a response that affects the B0 field.

Linear ECs consist of eddy currents that have an x, y, or z response. On-axis terms are linear ECs with the same stimulus and response, like x→x, y→y, or z→z. Crossterms are linear ECs with response on a different axis from the stimulus, like x→y, z→x, etc. B0 refers to any term with a B0 response, such as x→B0.

4.2.1.4.1.2 Time constant range

The range of time constants observed in the system covers roughly seven orders of magnitude, from 20 µs to 200 sec. While shorter time constant ECs may exist, the gradient field waveforms do not have enough high frequency content to stimulate these ECs. Similarly, ECs with time constants greater than 200 seconds become less significant, as they vary too slowly to be of significance to imaging. Generally, there is an inverse relationship between amplitude and time constant for observable ECs. This is because a small amplitude, short time constant EC is negligible when a larger, longer time constant EC is present.

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4.2.1.4.1.3 Eddy current measurement

The system's primary eddy current measurement technique is the *grafidy* tool. This tool drives a trapezoidal current pulse into the chosen gradient coil axis, and measures magnetic field during or immediately following the pulse. An array of six grafidy *probes* (sometimes called *coils*, and shown in Fig. 4–7) measures the field at six locations.

Each probe consists of a miniscule sealed MR sample inside a tiny RF coil. The field at the probe location is measured by creating an FID; the phase slope of the FID gives the FID frequency, allowing calculation of the local magnetic field. The sample is very small to give high spatial selectivity, and to limit the dephasing across the sample volume.

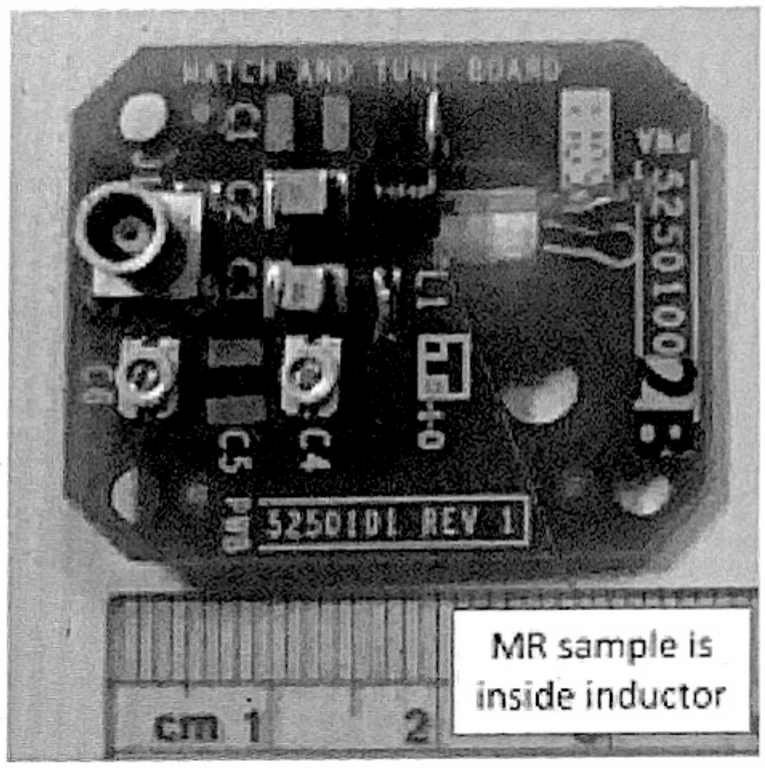

Fig. 4–7: A *grafidy* probe

4.2.1.4.1.4 Eddy current compensation

Compensation (not including HOEC compensation done for EPI sequences) includes addressing zero and first order eddy current fields. Compensation reduces the effects of these eddy currents to a small fraction of their uncompensated effects.

Linear eddy current compensation

Linear eddy current compensation is implemented by modifying the gradient current waveforms; preemphasis is applied to each waveform to offset the effects of the eddy currents. The present systems implement ECC with signal processing applied to the digital gradient waveforms; many other alternatives have been used in the past.

In general, linear ECCs give at least a 10X reduction: post-compensation measurements show 10X less residual EC field compared to pre-compensation measurements.

B0 eddy current compensation

B0 eddy current compensation is implemented by modifying the system's RF frequency; in essence, the frequency is modified to track the resulting changes in the B0 field.

4.2.1.4.2 B0 drift

This compensation applies to MR750w and Rio systems. The B0 drift tool characterizes changes in system center frequency (CF) caused by gradient activity so it may be compensated. One of the primary mechanisms causing CF drift is change in the passive shim material's temperature; this temperature change is caused by 1) gradient coil heating due to scanning, which warms the passive shim materials, and 2) eddy currents induced in the p-shim materials by gradient fields. The former cause is more significant than the latter. Note that although larger on 3TLC systems, other magnets exhibit B0 drift just not to a degree that compensation has been deemed to be required. Note that compensation on a Rio MR750 upgrade (with a G3 magnet) is being considered.

Characterization is done once as part of system installation by measuring the response (change in MR signal frequency) to a stimulus (gradient power on a per axis basis). A B0 drift tool (aka B0 drift cal) automates this characterization.

During system use, B0 drift compensation is always operating. The system monitors gradient coil currents, and predicts gradient coil power from these GC currents using parameters from a model to characterize resistance as a function of frequency. The GC power is then used to predict changes in B0 strength. The system's center frequency is changed to match the predicted change in B0 strength.

Performance specifications include the maximum MR signal frequency deviation from the nominal in a 10min timeframe.

B0 drift compensation uses the ABC model $(A + (B/10^5)*\log(f)^C)$ to characterize gradient coil resistance to estimate gradient power. This is true even for the Rio system that uses a more accurate 10-parameter ladder model to estimate power when setting up pulse sequence protocols. For the Rio system, the resistance at low frequencies was updated from what the ABC model estimates for DWI/DTI scans to account for overcompensation measured when using the ABC model. Refer to DOC2027980 for details.

4.2.1.4.3 HOEC

The HOEC (High Order Eddy Current) tool is a phantom-based measurement that characterizes the time dependent field errors (eddy currents). This tool is different from the *grafidy* tool in that instead of collecting only 6 points in space and characterizing the zero and first order eddy currents, the HOEC tool uses a large number of pixels in a phantom to characterize up to third order eddy currents. The current tool design is able to characterize eddy currents whose time constants range from approximately 5ms – 1000ms. The output of this tool is used with the EPI sequence to minimize the image quality impact by altering the waveforms (pre-emphasis) as well as when processing the images.

4.2.2 System safety

System safety limits (abbreviated to SSL in this section) define the safe operating space for a specific subsystem, component, or characteristic of the system. For example, SSLs exist for gradient coil power dissipation and RF amplifier output power, among others. The "system safety" term is used as these limits were originally put in place to protect the *system,* as opposed to the *patient*. In current systems, SSLs play a role in protecting both system and patient; for example, the SSL on gradient coil dissipation is part of the controls put in place to limit patient bore temperature.

System safety limits set the boundaries of what can and cannot be done with the subsystem, component, or characteristic (collectively called the "object"). There are three major aspects to a system safety limit.

1. The system safety limits *define the allowable use* of the object. A pulse sequence must operate within all system safety limits; a gradient waveform that would exceed the gradient coil power limits should not be run on the system.
2. The system safety limits define the capabilities needed of the object itself, and of related subsystems or components that it. Therefore, the system safety limits provide guidance for the design and testing of hardware. For example, the RF body coil must be able to create any sequence allowable

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within the B1 system safety limit; in addition, any receive only surface coils placed in the body coil transmit's field must tolerate the same sequences without failure.

3. The system safety limits define a test strategy for the object. For example, gradient coil power dissipation varies with pulse sequence type and imaging parameters. Since there are more different sequences than can be reasonably tested, the system safety limits are used to define a unified test strategy that covers worst-case allowable use of the object.

System safety limits exist for B1 fields, RF amplifier capability, gradient coil power dissipation, and gradient driver capability.

4.2.2.1 B1 related system safety limits

The *B1* related system safety limits are in place to limit the *B1* field. The transmit *B1* field induces voltages on items exposed to the field; on conductive items, currents will be induced as well. In turn, this leads to heat dissipation. The *B1* related system safety limits are there to place limits on the induced voltage, current, and heating.

Since B_1 strength is affected with the transmit power setting and the object in the transmit coil, it can be difficult to measure directly. GEHC MR uses an MR based measurement, specifically the transmit gain setting chosen by auto-prescan, as a reference point for B_1 strength. See section 3.3.2, *Interaction with* $\mathbf{B_1}$, for more information.

The *B1* related SSLs are only "enforced" by the design of the pulse sequences; the system has no direct *B1* monitoring, and therefore no direct limit enforcement. The pulse sequence designs are allowed to run right up to the *B1* system safety limits: if a B1rms limit of 3.6uT is applied, sequences may be designed to run at 3.6uT rms. However, the actual *B1* during the scan will vary with transmit gain (TG) settings, which are determined by prescan. Therefore, the *actual B1* may be higher or lower than the *expected B1*; this variability should be considered when testing for *B1* related effects.

The SSLs depend several factors. First, different transmit coils may have different limits; the body coil limit may be different than the head coil, as the overall volume is larger, and head coils generally do not have receive only coils inside them. Second, individual receive coils may need different SSL values; one particular receive coil design may need lower *B1* limits than another. When a receive-only coil is used, the transmit coil may have different *B1* SSLs than the receive coil; in this case, *the lowest limit must be observed*. If the transmit coil has a B1rms limit of 3.6uT, and the receive coil has 3.0uT, the pulse sequence must observe the lower 3.0 limit.

The values for the *B1* related SSLs are kept as part of the RF coil database.

4.2.2.1.1 B1 peak

B1 peak (or the alternative phrase *peak B1*) is defined as the highest value of *B1* used in the pulse sequence. The primary purpose is to limit the peak voltage and current that will be induced on an object.

$$B1peak = \max(B1(t)) \tag{4.1}$$

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For the body coil, peak *B1* of 22-25uT are typical; this is controlled to prevent overexpose of removable or receive-only coils. For other volume coils, such as a head or knee coil, higher peak *B1* values of up to 50-60uT are typical.

4.2.2.1.2 B1 rms

B1 rms is defined as the rms value of *B1* used in the pulse sequence. The primary purpose is to limit the heat that will be induced in or on an object.

$$B1_{rms} = \sqrt{\frac{1}{TR}\int_0^{TR} B1^2(t)dt} \qquad (4.2)$$

4.2.2.1.3 Integrated B1 squared

The *integrated B1 squared* limits controls the maximum "B1 energy" used in the pulse sequence. The primary purpose is to limit the energy deposited into an object by a single pulse; this limits the thermal shock that can be caused by a single RF pulse.

$$B1rms = \int_0^{PW} B1^2(t)\,dt \qquad (4.3)$$

4.2.2.1.4 TG cap

"Hard" RF pulses (non-bandwidth limited rectangular pulses) have a step function on the pulse's leading edge. If the RF amplifier's step response has overshoot, the resulting transient on high power pulses will expose the transmit chain to power above the expected full power level. Because high voltage failures of the body coil pose an unacceptable patient risk, a method to prevent these high power transients was developed.

There are three conditions needed to create a transient overshoot that exceed the expected full power:

1) use of a hard pulse,
2) an amplifier with overshoot in the step response, and
3) a TG setting high enough that the transient exceed rated power.

The prevention method takes all three of these factor into accounts. For any pulse sequences that use hard pulses, on systems using amplifier designs known to have overshoot, TG will be limited to 170 (-3dB relative to full power). This ensures that any hard pulses will be limited to half power; so the overshoot will not exceed full power.

The only amplifier known to have significant overshoot is the 1.5T Analogic 8120.

4.2.2.2 RF amplifier capability

RF amplifier capability models were developed to indicate whether a particular amplifier design should be able to complete a particular pulse sequence, and are used to ensure that pulse sequences do not exceed the capabilities of an amplifier.

Up till about 2000, RF amplifiers were specified to deliver a fixed amount of average power; however, the amplifier was *assumed* to be operating at full power, where it was most efficient. The average power

available at less than full power was not specified. Since almost all pulse sequences use shaped RF pulses, there was no reliable method for calculating the usable capabilities of the amplifier.

The current RF amp capability models use a "stress function", called *f(t)*, that models the stress on the amplifier as a function of RF voltage, and a "windowed average of *f(t)*", called *F(T)*, to determine if à pulse sequence will overstress the amplifier. The form of the stress function and windowed averaging curves are kept general, to allow a wide variety of RF amplifier designs to be used with minimal changes to the system's software.

The RF amplifier capability SSL is written to force the "long term" average stress on the amplifier to be less than 100%. For shorter averaging windows, the stress may be greater than 100%, provided that lower stress operation follows to hold the long term average to 100%.

4.2.2.2.1 Stress function definition

The *stress function f(t)* is designed to model the stress, in a general sense, on an RF amplifier.

$$f(t) = g(t)\left[K_U + K_V \cdot |v(t)| + K_P \cdot |v(t)|^2\right] \tag{4.4}$$

where

K_U = unblanking coefficient
K_V = RF voltage coefficient
K_P = second order RF voltage coefficient, proportional to RF power (optional)
$g(t)$ = RF unblank gating signal
$v(t)$ = RF envelope voltage, normalized to ±1.00 at rated output power

This stress function is not intended to directly correspond to any one of the many parameters that may limit the capabilities of an RF amplifier (tube temperature, FET bias, power supply voltage, AC line current, etc.) Instead, it reflects the general conditions that all pulsed amplifiers share: unblanking the amplifier increases stress on the amp, and producing RF power increases stress, with higher power producing more stress.

The overall scaling of the coefficients is chosen so that long term time-averaged value of *f(t)* is 1.00, or 100%, when the amplifier is running at full capability.

Using a general model, while allowing for different K_U, K_V, and K_P coefficients, allows the model to be applied to a variety of amplifier designs and technologies. This simplifies the model's implementation in the MR system software.

The form of the stress function – up to a 2nd order polynomial on *v(t)* – was chosen to ease calculations in older systems. The RF pulse definition software structure had pre-calculated, stored values for the pulse width, absolute area, and energy of each pulse profile. These could be used to calculate an equivalent *f(t)* using rectangular waveforms, which reduced the integration needed in the windowed averaging process (see below) to a summation.

Example: a typical stress function *f(t)* might be

$$f(t) = g(t)\left(0.5 + 12 \cdot |v(t)|\right)$$

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The figure below shows *f(t)* resulting from a three-pulse RF sequence. The second and third pulses are full power pulses, so they have *v(t)* = 1 and *f(t)* = 12.5. The "pedestal" resulting from the unblank gating is seen under all three pulses.

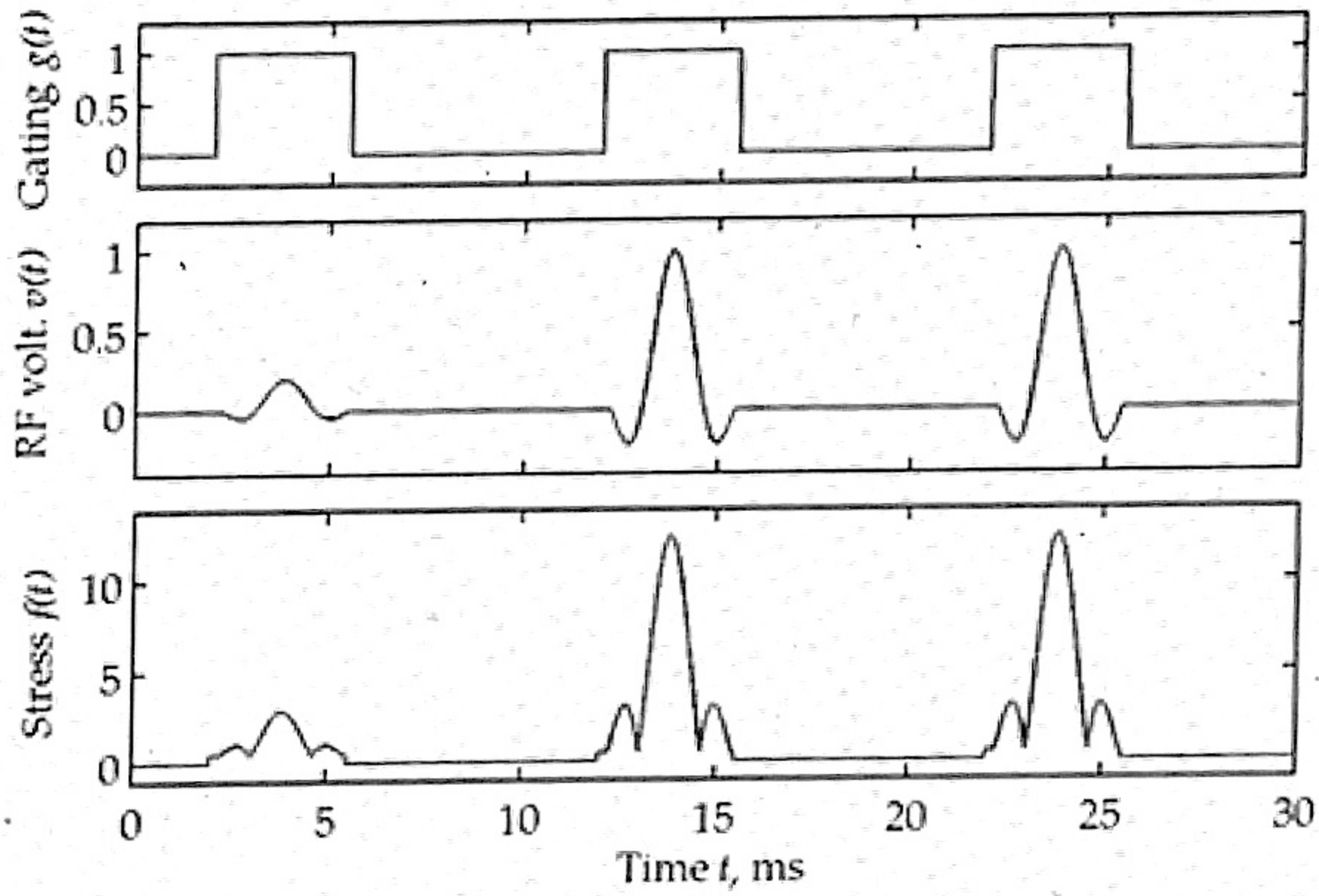

Fig. 4-8: Typical stress function

4.2.2.2.2 Windowed average curves

To determine if a particular sequence can be run on a particular amplifier, the resulting *f(t)* curves are time averaged. The allowable stress depends on the length of the averaging window; in general, higher stresses are allowed for shorter interval, while the long-term average must be limited to 100% of the amplifier's capability, or 1.00. An averaged *f(t)* over a window length *T*, or *F(T)*, is defined as

$$F(T) = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)\,dt \qquad (4.5)$$

This averaged stress is limited by curves that define the allowable stress at any window length *T*, which set the upper limit on *F(T)*,

$$F(T) < \text{minimum}\begin{cases} A_2 \\ A_1/T + F_1 \\ A_0/T + F_0 \end{cases} \qquad (4.6)$$

where

T = the length of the averaging window

F(T) = the average value of *f(t)* over window width *T*